IT Support Ticketing System Analysis

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing IT support ticketing raw data using Excel and creating an interactive Power BI dashboard to derive meaningful business insights.

The main goal of this project is to clean and organize raw IT ticket data so we can measure IT performance, technician work, equipment reliability, costs, and user satisfaction to help managers make better decisions.

2. Data Sources

- **Primary Source:** IT Support Ticketing System Analysis (Raw Dataset, 500 records, 30 attributes).
- **Data Timeline:** 2023 to 2026 operational records.
- **Domain:** IT Service Desk and Infrastructure Management.

3. Problem Statement

- **Operational Bottlenecks:** Find out which IT problems happen most often, which urgent issues need immediate attention, and where delays are happening on campus.
- **Technician Productivity:** Evaluate technician efficiency by comparing estimated vs. actual resolution hours and measuring overall resolution completion rates.
- **Asset Management:** Track vendor reliability, asset maintenance cycles, software licensing, and hardware failure rates to optimize IT procurement.
- **Service Satisfaction:** Check customer satisfaction ratings by department, user role, and issue severity to make IT support better.

4. Attribute (Column/Features) Details

Attribute Name	Description
Ticket_id	Unique identifier for each IT support ticket (e.g., ITK-00001)
Requester_name	Full name of the user
Requester_email	Email address of the requester
Department	Academic or administrative department (e.g., Computer Science)
Ticket_category	Classification of issue (e.g., Hardware Fault, Network)
Priority_level	Urgency level of ticket (Low, Medium, High, Critical)

Ticket_status	Current state of ticket (Open, In Progress, Resolved)
Assigned_to	IT support staff member assigned to resolve the issue
Created_at	Timestamp when the ticket was logged
Resolved_at	Timestamp when the ticket was resolved
Asset_tag	Unique hardware/device identifier (e.g., AST-000001)
Location	Physical campus location or lab block
Description	Detailed issue narrative reported by user
Resolution_notes	Technical actions taken to resolve the ticket
Is_escalated	Indicates if the ticket required higher-level intervention
Estimated_hours	Estimated time required for resolution
Actual_hours	Actual time spent resolving the issue
Cost_center	Financial account code billed for support
Contact_phone	Phone contact number of the requester
User_role	Role of requester (Faculty, Student, Staff)
Severity_score	Calculated impact score (1 to 10 scale)
Vendor_name	Hardware/software provider or manufacturer
Warranty_expiry	Expiration date for hardware/software warranty
Last_maintenance	Date of last scheduled preventative maintenance
IP_address	Assigned network IP address
Mac_address	Physical hardware network adapter address
Os_version	Operating system and version installed on asset
License_key	Software product license key code
Follow_up_required	Flag for required post-resolution follow-up

Satisfaction_rating	Rating submitted by requester after closure

5. Tools and Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-processing (Excel/Power Query)

Tasks Performed

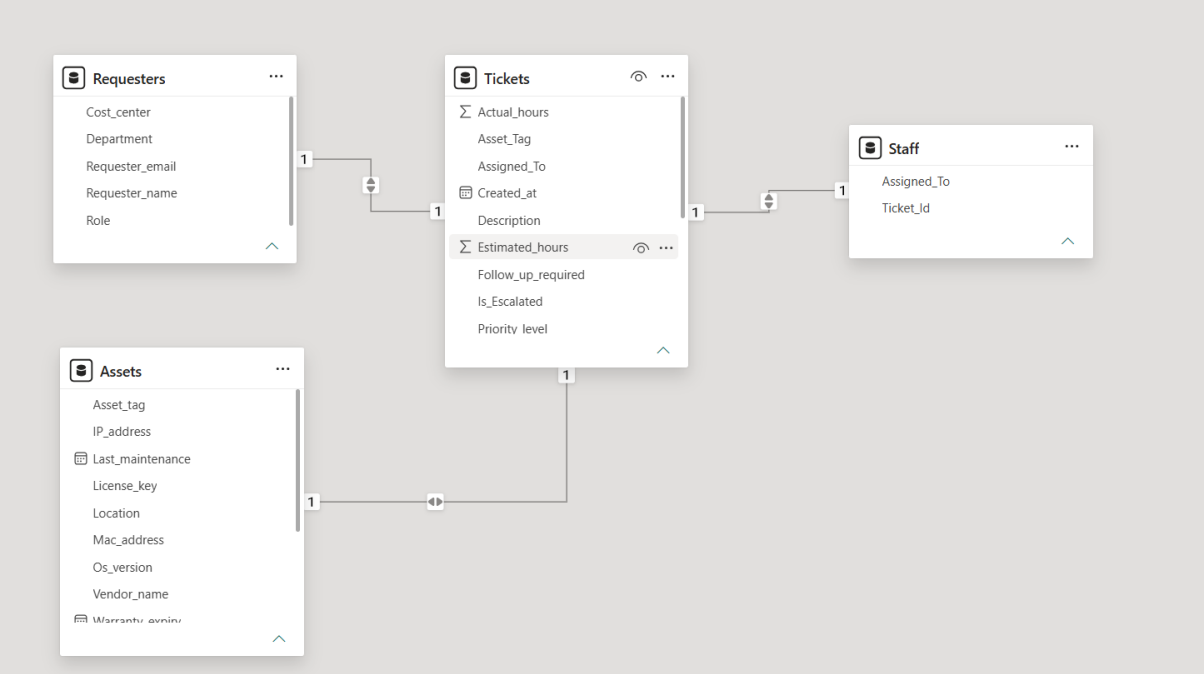
- **Data Cleaning & Transformation:** Checked null values and verified text formatting across 500 rows and 30 columns.
- **Standardization:** Extracted numerical metrics and standardized date formats.
- **Relational Normalization:** Relational Star Schema structure to remove redundancy.

Relational Data Decomposition (Fact and Dimension Tables)

- **Tickets Fact Table ('tickets'):** ticket_id, requester_email, asset_tag, assigned_to (FK), ticket_category, priority_level, ticket_status, created_at, resolved_at, description, resolution_notes, is_escalated, estimated_hours, actual_hours, severity_score, follow_up_required, satisfaction_rating.
- **Requesters Dimension (`requesters`):** requester_email, requester_name, department, user_role, contact_phone, cost_center.
- **Assets Dimension ('assets'):** asset_tag (PK), location, vendor_name, warranty_expiry, last_maintenance, ip_address, mac_address, os_version, license_key.
- **Staff Dimension (`staff`):** ticket_id, assigned_to.

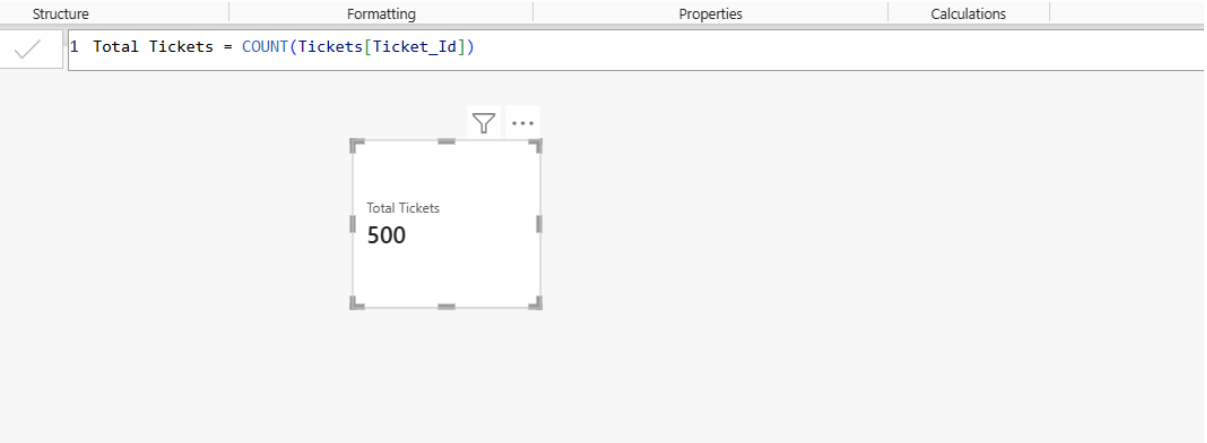
7. Data Modelling and DAX (Power BI)

Data Model

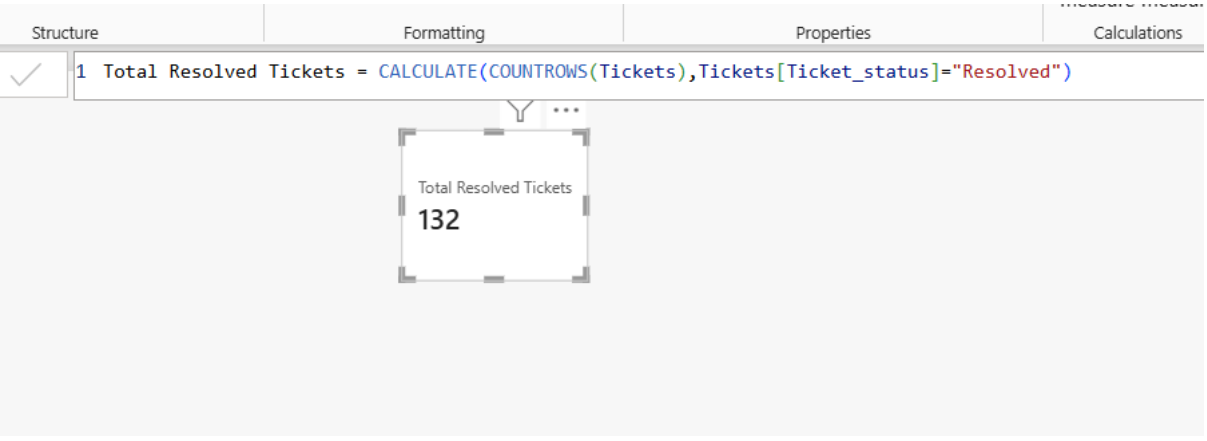


Dax Measures

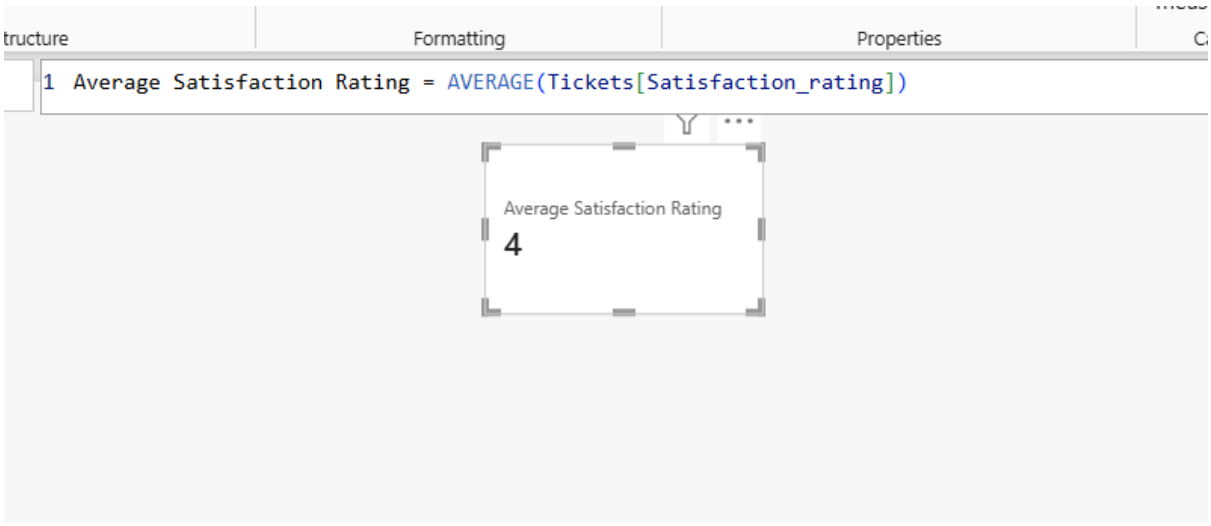
Total Tickets



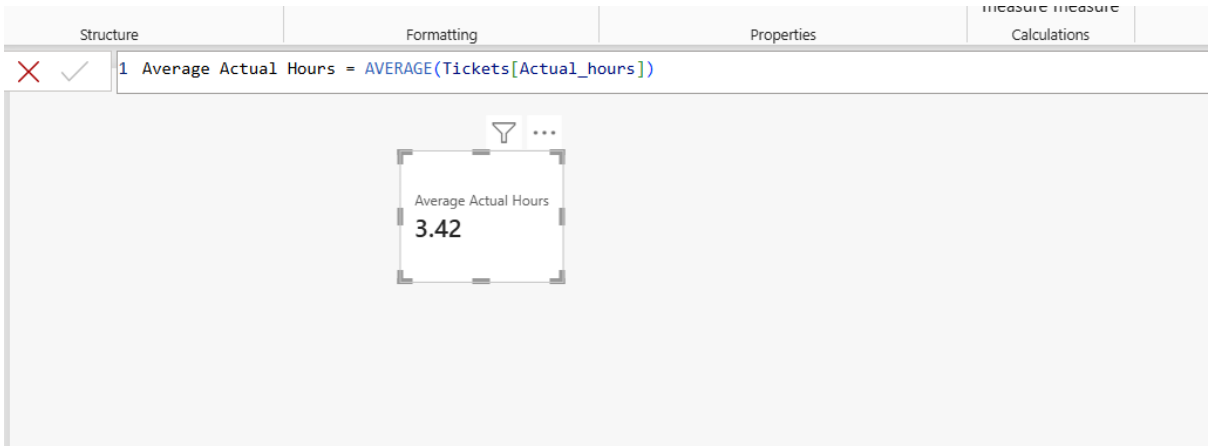
Total Resolved Tickets



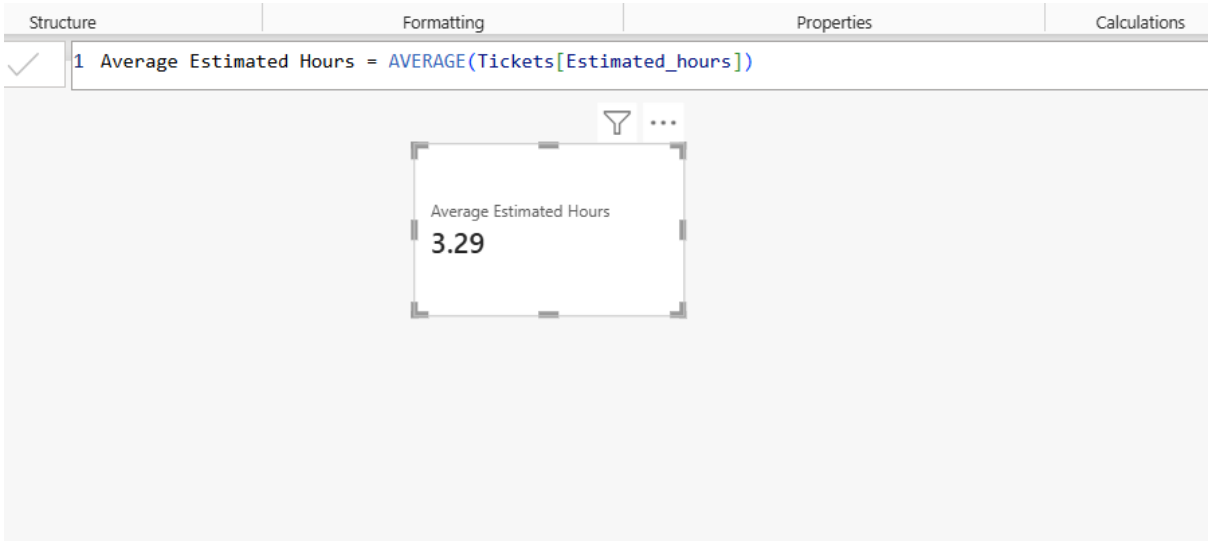
Average Rating



Average Actual Hours

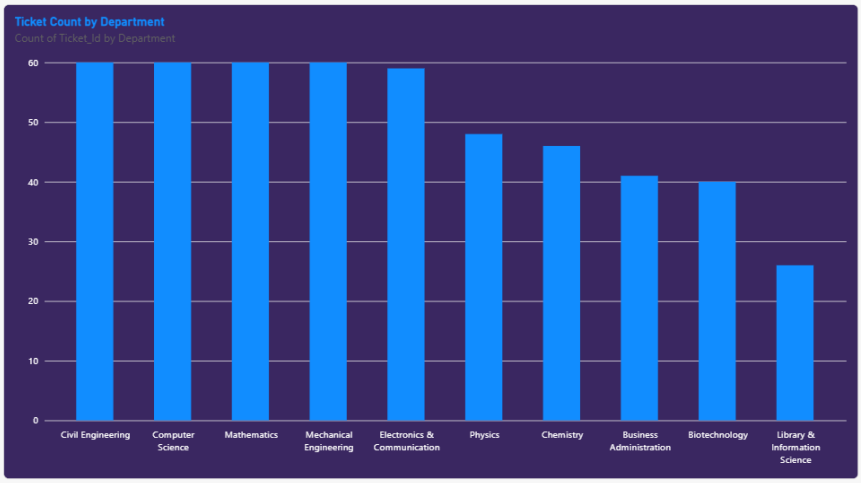


Average Estimated Hours

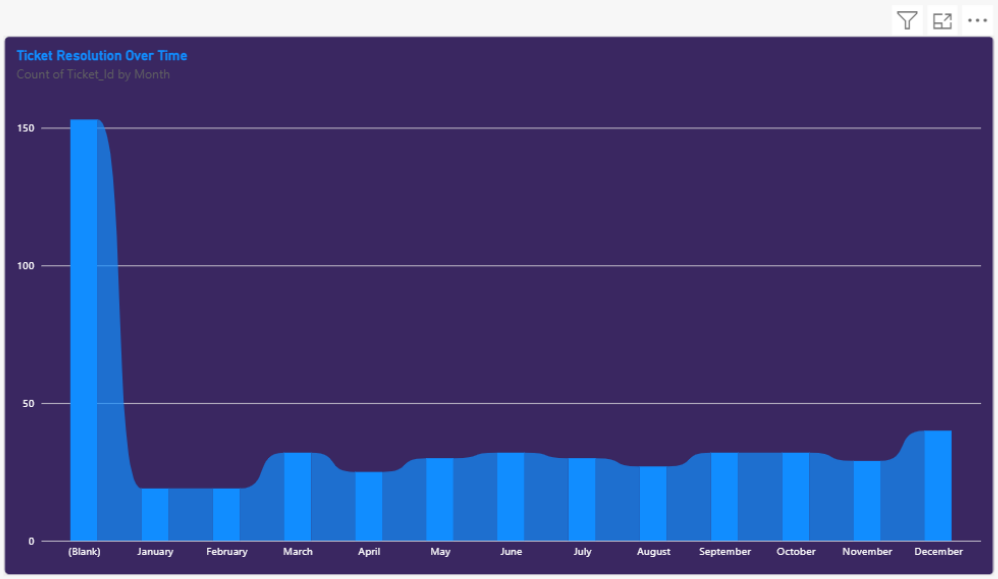


8. Analysis and Visualizations (Power BI)

Ticket Count by Department



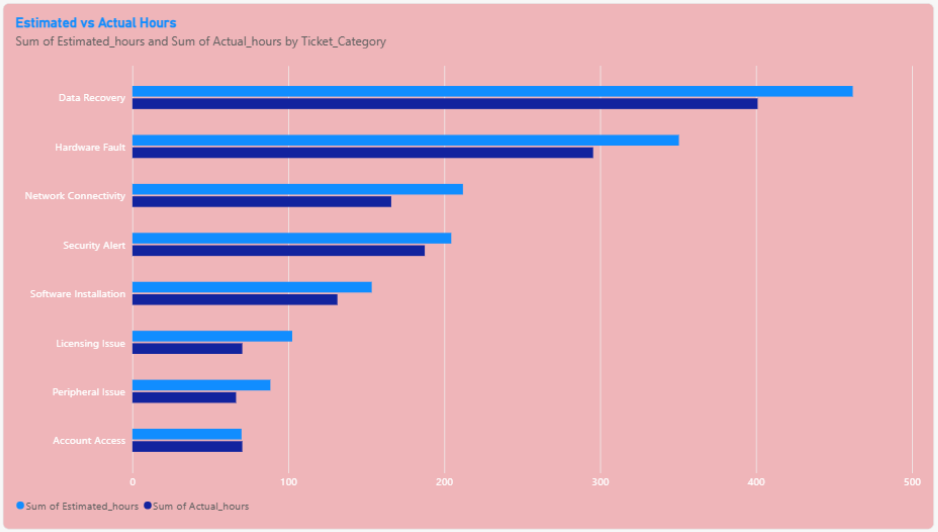
Ticket Resolution Time



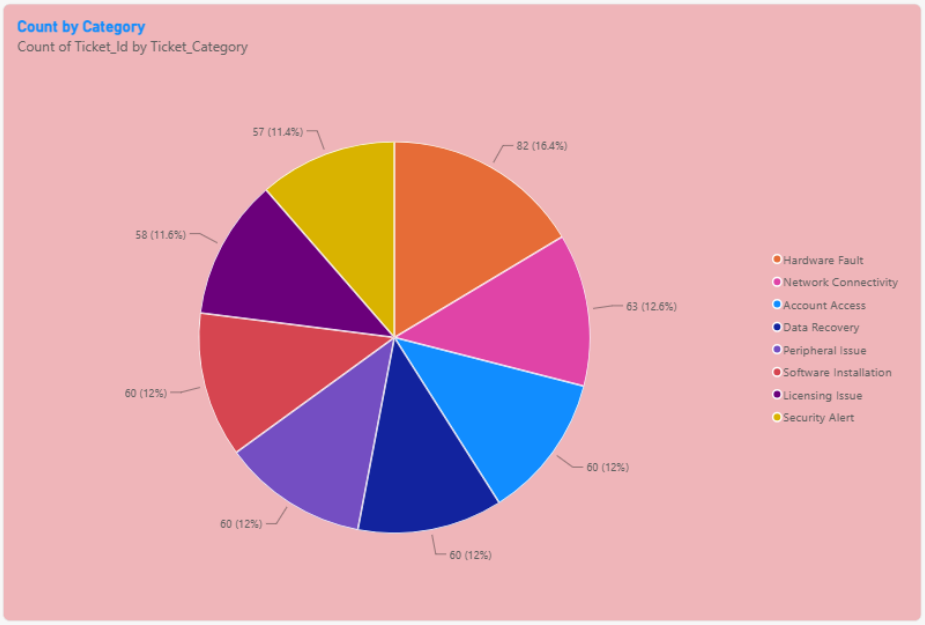
Ticket Count by Vendor



Estimated vs Actual Hours



Count by Category

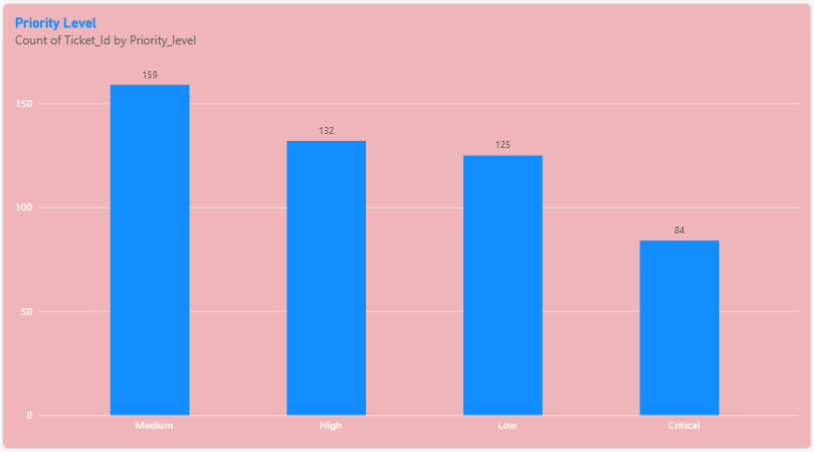


Staff Workload

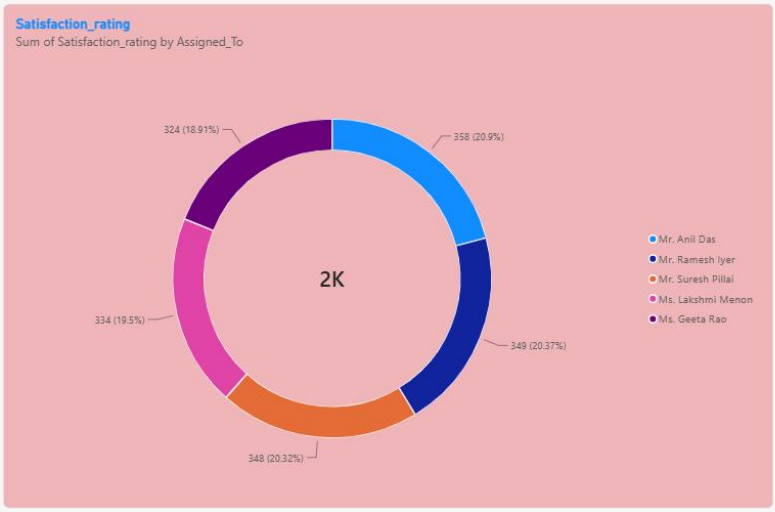
Staff Workload Summary

Assigned_To	Closed	In Progress	Open	Resolved	Waiting on User	**Total**
Mr. Anil Das	19	18	13	31	19	**100**
Mr. Ramesh Iyer	12	30	25	23	10	**100**
Mr. Suresh Pillai	12	19	22	37	10	**100**
Ms. Geeta Rao	10	20	19	19	32	**100**
Ms. Lakshmi Menon	9	35	25	22	9	**100**
Total	**62**	**122**	**104**	**132**	**80**	**500**

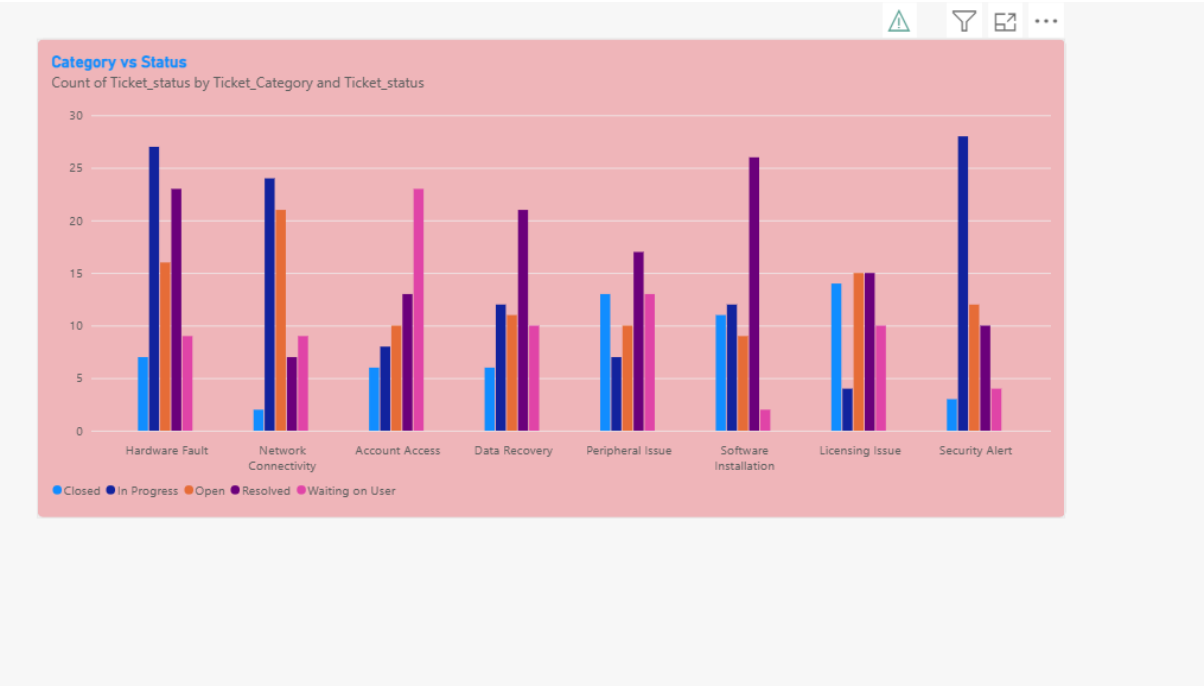
Priority Level



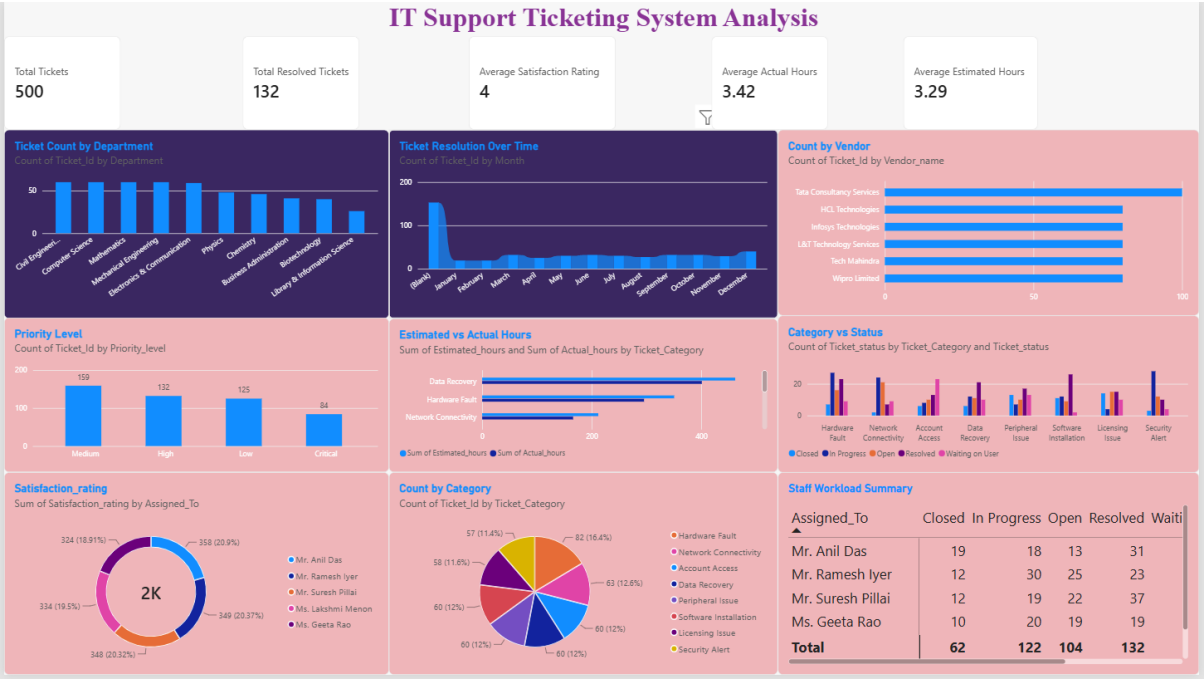
Satisfaction Rating



Category vs Status



Dashboard



9. Insights

Descriptive analysis

- A total of 500 tickets covering 8 different problem types, with Hardware Faults (82 tickets) and Network Problems (63 tickets) being the most common issues.
- On average, it took 3.42 hours to resolve a ticket, which was very close to our initial estimate of 3.29 hours.
- Overall user satisfaction averaged 4 out of 5.

Diagnostic analysis

- The small gap between the estimated time (3.29 hours) and actual fix time (3.42 hours) happened because complex tasks like hardware checks and network rewiring issues that needed extra repair time.
- Despite these minor delays, overall user satisfaction remained strong at 4 out of 5 because technicians met their response targets and fixed standard tickets quickly before they caused major disruptions.

Predictive analysis

- Urgent repairs will likely keep taking 10% to 15% longer than planned unless we add extra time to our schedules.
- Departments dealing with frequent network drops may give lower satisfaction scores in the future.
- Additionally, technicians handling complex asset repairs will likely continue to face heavy workloads and higher working hours per ticket, pointing to staff shortages that could cause future delays if tasks are not rebalanced.

Prescriptive analysis

- To improve future performance, we should add a small time buffer to our initial estimates for difficult tasks like data recovery to make scheduling more accurate.
- Set up dedicated support for departments with lower satisfaction scores to resolve their issues faster.

10. Conclusions

- Overall, the IT support team performed well, fixing 500 tickets with an average resolution time of 3.42 hours per ticket (very close to our 3.29-hour estimate) and earning an average user satisfaction score of 4 out of 5. While everyday hardware and network issues were fixed quickly, complex tasks like Data Recovery caused some delays.
- By adding extra time buffers for hard tasks, and giving extra help to lower-scoring departments, the IT team can prevent staff burnout, improve repair schedules, and keep satisfaction high across campus.